1) 0,580 g C_xH_y

0,480 g C 0,100 g H

$\frac{0,480}{12} = 0,04$ mol C $\frac{0,100}{1} = 0,10$ mol H

$\frac{0,04}{0,04} = 1$ → 2 $\frac{0,10}{0,04} = 2,5$ → 5

C_2H_5 (Kuba) → 29 g/mol

(33,6 mL'si 0,087 g

22400 mL'si x g) x2

x = 58 g/mol → $(C_2H_5)2$ · C_4H_{10} (Molekül f.)

3) $\left(\frac{v_1}{v_2} = \sqrt{\frac{M_2}{M_1}}\right)$ $M_{A(H_2)} = 4$ g/mol

$M_{A(O_2)} = 16 \cdot 2 = 32$ g/mol

$\frac{v_{H_2}}{v_{O_2}} = \sqrt{\frac{32}{4}} = \sqrt{8} = 2\sqrt{2}$

$\frac{v_{H_2}}{v_{O_2}} = \frac{4 \text{ mol/dk}}{v_{O_2}} = 2\sqrt{2} \rightarrow \frac{4}{2\sqrt{2}} = v_{O_2} = \sqrt{2}$ mol/dk

4) $P \cdot V = n R T \rightarrow V = \frac{nRT}{P}$ $\frac{V}{n} = \frac{RT}{P}$

5) $CO_2 + H_2O \longrightarrow C_2H_4 + 3O_2$ $+330$ kkal

$2C + 2O_2 \longrightarrow 2CO_2$ -188

$H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$ -58

+

$330 - 188 - 58 = 84$ kkal

7) ($Q_{alınan} = -Q_{verilen}$)

($m.c.\Delta t = m.c.\Delta t$)

$c_{su} = 4,18$ j/g°C $= 1$ cal/g°C

İzole kapta enerji kaybolmaz.

$50.4,18.(28,8-22) = 150.c_{kurşun}.(100-28,8)$

$c_{kurşun} \downarrow$

$0,13$ j/g°C

8) $CaCO_{3(k)} \longrightarrow CaO_{(k)} + CO_{2(g)}$

($\Delta H_{Tepkime} = \Delta H_{Ürünler} - \Delta H_{Girenler}$)

$((-638)+(-396)) - (-1207) = 173$ kj/mol (+ endotermik ısı gerekli)

9) $1s\,2s\,2p\,3s\,3p\,4s\,3d\,4p\,5s\,4d\,5p\,6s\,4f\,5d\,6p\,7s$ ...



n=3 enerji düzeyinde $m_s = 1/2$ olan e^- sayısı $= 1+3+5 = 9$

12) $1s\ 2s\ 2p\ 3s\ 3p\ 4s\ 3d\ 4p$

⊗ ⊗ ⊗ ⊗ ⊗ ⊗ ⊗

⊗ ⊗ ⊗

⊗ ⊗ ⊗

⊗

⊗

6 elektron

D- 3s

15) XO_3^{-2} toplam e^- : 32 X per cet yeri = ? (O : 8)

$X + 24 + 2 = 32$

$X = 6$

${}_6X : 1s^2 2s^2 2p^2$

2 per 4A

16) $1s^2 2s^2 2p^6 4s^1$ X

$4s^2$ Y ← en çok proton olan, en güçlü seker, çap az

$2p^4$ T

$2p^5$ U

$2p^6$ Z

17) proton → Ca > K > Ar > Cl > S

ters

çap → S > Cl > Ar > K > Ca

18) 1 e^- kaybında atom soygaz halne dönüşüyorsa 2. yi koparmak zorlaşır

$Na: 2)8)1) \rightarrow Na^{+1} \rightarrow 2)8)$

soygaz

19) pe tablo → İyonlaşma enerjisi artar
(Basamağı 5 yukarı)

27) CH_4 : 12+4 : 16 g/mol

$\frac{6,07}{16} = 0,379$

29) $2Pb(NO_3)_2 \rightarrow 2PbO + 4NO_2 + 1O_2$

4N
12O

30) $M_A \cdot V_{Asit} = M_B \cdot V_{Baz}$

$0,295 \cdot 25 = 0,0738 \cdot V_B$
↳ 100 ml